

Chemical Library
65 856 African NP-
inspired molecules

Target Panel
PfDHFR & PfCRT
WT + 6 resistance mutants

Docking-derived estimand

Set-C Docking Cohort
17 candidates $\times$ 8
states = 136 systems
MPO $\geq$ 0.70, SYBA $>$ 0, SI $>$ 10

AutoDock Vina 1.2.7
Exhaustiveness 64, V2-corrected grids
→ Docking scores per complex

Docking-RRS (primary)
5-class retention: A*, A, B, C, D
+ ACSI + PNS ranking

MD-derived estimand

Parent MD Cohort
4 WT complexes (non-overlapping)
10 ns each, CHARMM36m/GAFF2

MD Trajectory Analysis
RMSD, contacts, H-bonds, bound QC
GB^{OBC2} MM-GBSA endpoint

MD-RRS (secondary)
Min-distance ratio per mutant
MMG-RRS ratio, bound fraction

Set-C MD Pilot
16 systems (PP-01, PP-02), 10 ns each
16/16 QC PASS, single replicate

Estimand Comparison
7/8 directional disagreements
⇒ non-equivalent

